

COMS W4156 Advanced Software Engineering (ASE)

November 16, 2023



Agenda

1. Introduction to Delivery, Deployment and DevOps
2. GCP Deployment with demo: Varsha Kanmuri (TA) and Griffin Newbold from 'TheJavaEngineers' (student taking class)
3. DevOps Challenges: Tsai-Chen Hsieh (student taking this class, but his presentation is from his midterm paper in my 6156 section in spring 2023)

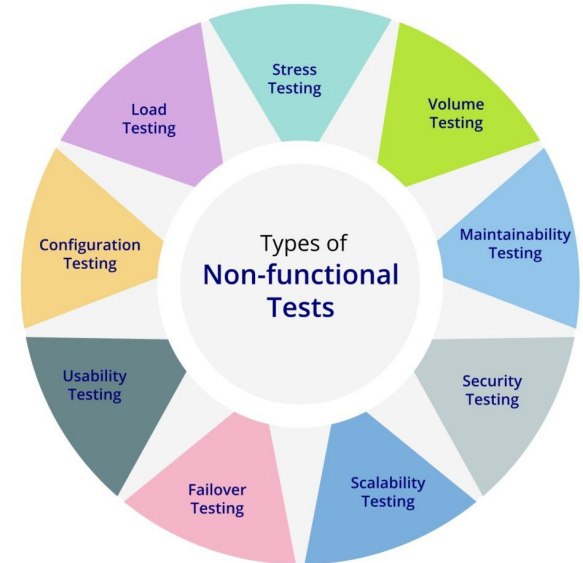
	Continuous Delivery	Continuous Deployment
Definition	A software engineering practice where code changes are automatically prepared for deployment.	A software engineering practice that allows code changes to automatically release into the production environment.
Benefits	- Improved developer productivity - Complete automation of the delivery process - Bugs are found and addressed earlier - New updates released more frequently	- Faster release process - No need for human intervention - Fixing problems is easier - Frequent updates to the customer.
Challenges	- Requires a strong CI foundation - Requires a good testing suite - Needs an automated deployment process - Features flags and their proper usage	- Requires an excellent testing suite - Documentation must remain up-to-date - Feature flags become an inherent part of the DevOps process.

CI/CD

Recall CI = Continuous Integration, which puts great emphasis on testing automation to check that the application is not broken whenever new commits are integrated into the main branch

CD = Continuous Delivery or Continuous Deployment extends CI, usually applied only when all CI tests and checks fully pass

In CI/CD context, needs to include non-functional as well as functional testing. ? Why ?

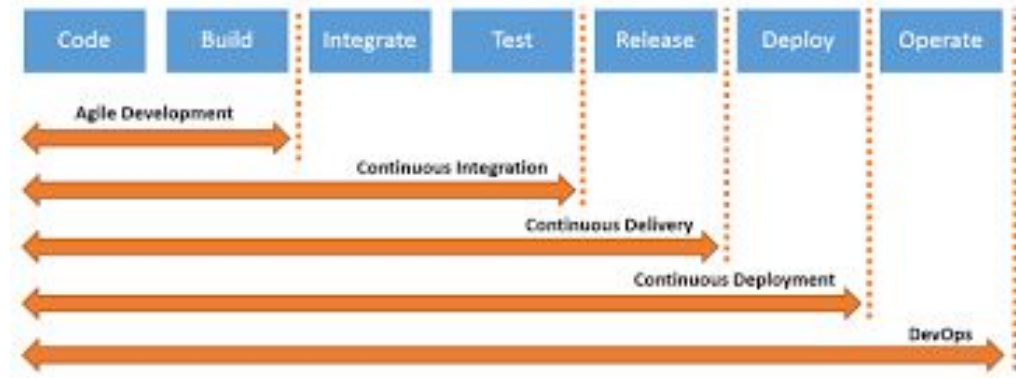


Delivery vs. Deployment

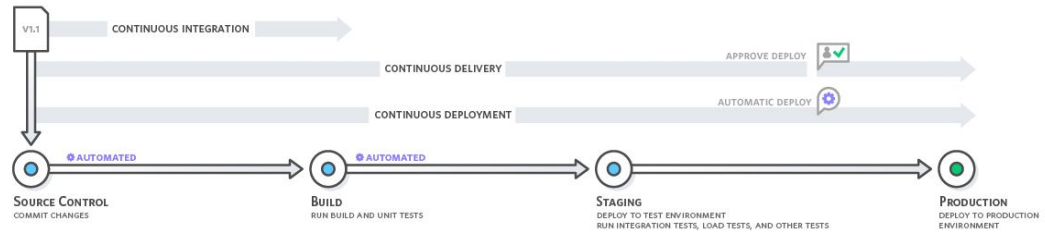
Delivery is process of preparing software for installation or update by customer on customer equipment (or potentially in cloud space rented or otherwise acquired by the customer) - previously 'shrinkwrapped', now typically downloaded

Deployment is process of preparing software for installation or update by in-house operations staff on in-house or cloud equipment for use by in-house and/or external users

Sometimes delivery vs. deployment terms are used interchangeably



What is DevOps?



Integration of in-house developers (dev) and in-house operations (ops) staff

Sometimes refers to a close feedback loop between vendor's developers and customers' operations staff

CD usually installs/updates software to staging environment for further testing rather than directly to production

Staging environment = scaled down replica of production environment with testers and maybe alpha users, but no regular users

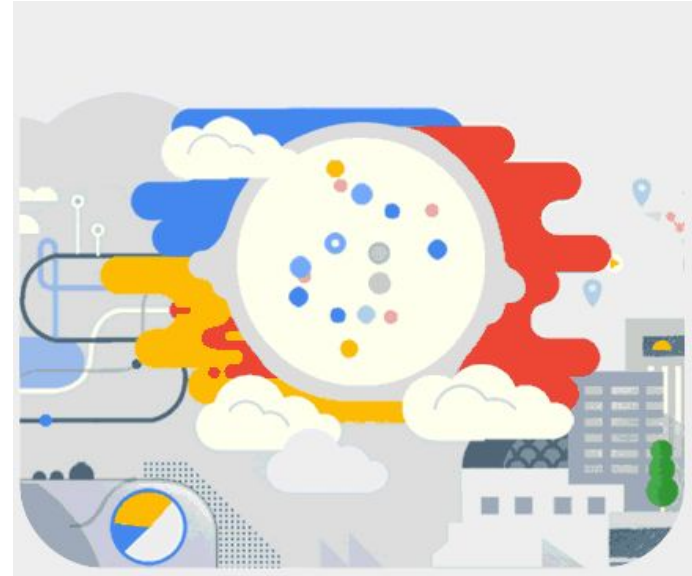
Often live monitoring and 'in vivo' testing in both staging/production environments

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GCP Deployment: Varsha Kanmuri and Griffin Newbold



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DevOps Challenges: Tsai-Chen Hsieh



Upcoming Assignments

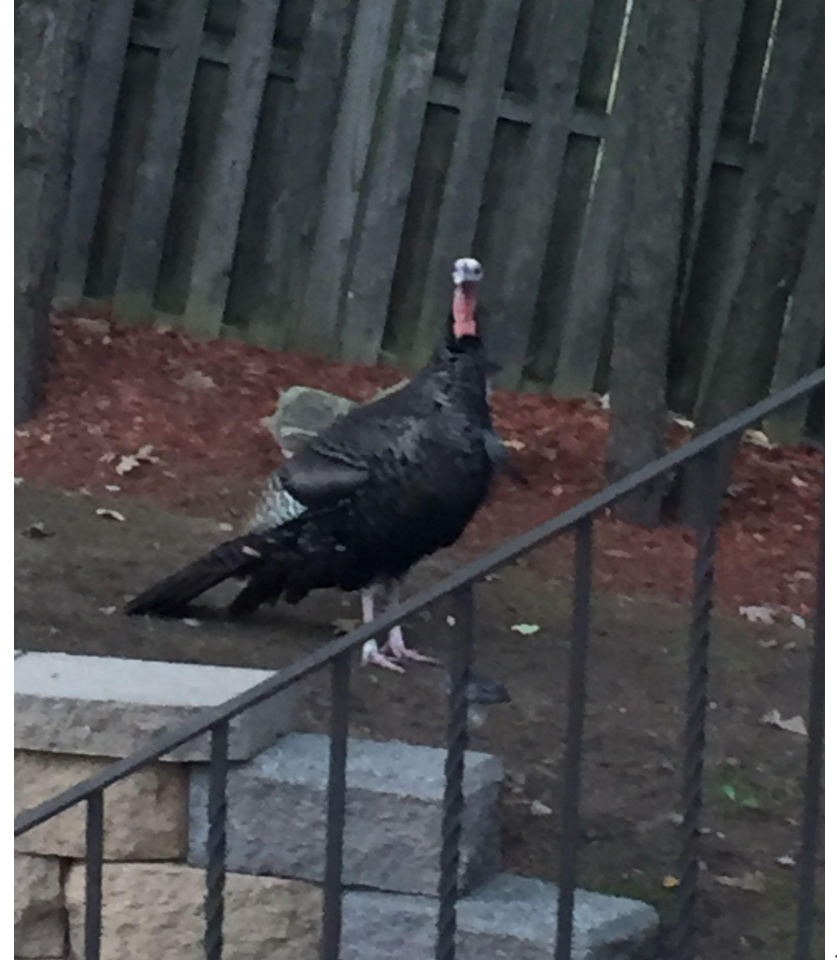
- [Second Iteration](#) due Thursday, November 30, by 11:59pm
- [Second Iteration Demo](#) due Thursday, December 7, by 11:59pm



Next Week

No class Thursday = Thanksgiving

? Should we have class on Tuesday ?



Ask Me Anything